

EDUCATION

Caldwell University, Caldwell, NJ

May 2027

Major: Computer Science | **Double Minor:** Mathematics, Business Analytics

- Dean's List, Honors Award Recipient, Presidential Scholarship
 - Courses: Data Structures & Algorithms, Web Development, Database Management, Calculus, Artificial Intelligence
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RESEARCH EXPERIENCE

Research Assistant | Caldwell University (Advisor: Prof. Bella Veksler)

Sept 2026- Present

Project: Do Large Language Models Know When Their Working Memory Fails?

- Investigating LLM working-memory performance and metacognitive awareness using verbal and spatial n-back tasks.
- Reviewing research in cognitive science and machine learning to identify limitations in existing LLM working-memory evaluations.
- Designing experiments to evaluate accuracy, confidence calibration, and performance across increasing memory demands.
- Developing an evaluation framework using Expected Calibration Error (ECE) and Brier Score.

Undergraduate Research Assistant | Caldwell University (Advisor: Prof. Vlad Veksler)

Sep 2025 – Aug 2026

Project: Scaling Long-Range Retrieval: A Comparison of LSTM, Transformer, and Mamba on a Needle-in-a-Haystack Task

- Developed a synthetic Needle-in-a-Haystack task to compare the long-range retrieval abilities of LSTM, Transformer, and Mamba.
 - Evaluated model performance across sequence lengths ranging from 32 to 1,024 tokens using PyTorch
 - Observed a decline in Transformer retrieval accuracy from approximately 99% at shorter sequences to 19% at 1,024 tokens.
 - Compared retrieval accuracy, training time, and memory usage to analyze performance and computational efficiency.
 - Poster accepted at CCSC Eastern Conference 2026, October 9–10, 2026.
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PRESENTATION

CCSC Eastern's 42nd Annual Regional Conference

Oct 2026

Scaling Long-Range Retrieval: A Comparison of LSTM, Transformer, and Mamba on a Needle-in-a-Haystack Task

Caldwell University Annual Research and Creative Arts Day

May 2026

Scaling Long-Range Retrieval: A Comparison of LSTM, Transformer, and Mamba on a Needle-in-a-Haystack Task

WORK AND LEADERSHIP EXPERIENCE

Data Analyst Intern, | Office of Institutional Research | Caldwell University | Caldwell, NJ

May 2026 -Aug 2026

- Built interactive dashboards and data visualizations to communicate research findings to faculty, staff, and administration
- Developed and launched surveys using IPEDSR database, including data collection, validation, and analysis
- Cleaning, organizing, and analyzing institutional datasets using R
- Conducting statistical analysis and data mining to extract actionable insights from complex datasets

NJAS Judging Website, Maintenance & Feature Update | *Web Development, Collaborative Fixes*

April 2026

- Took over the New Jersey Academy of Science (NJAS) judging website after the original developer was unavailable.
- partnered with three peers and the NJAS organizing team to deliver required changes ahead of the competition cycle.
- Modified site functionality to align with updated judging criteria and user-flow requirements, coordinating directly with stakeholders to validate scope and acceptance criteria.
- Debugged and patched existing code to ensure reliable performance during active judging, and documented changes to streamline future handoffs.

Teaching Assistant, CS 196: Computer Programming II, Caldwell University

Jan 2025 – May 2025

- Assisted the instructor in assignment design, and grading coursework for 50+ students in Python programming, covering object-oriented programming (OOP), recursion, and python libraries.
- Led weekly tutoring sessions to help students understand programming concepts and prepare for assignments, quizzes, and exams

- Provided hands-on guidance during programming labs, helping students debug code and develop problem-solving skills.

Teaching Assistant, CS 302: Design and Analysis of Algorithms, Caldwell University **August 2025 – Dec 2025**

- Provided academic support to 15 students in algorithm design, graph theory, complexity analysis, debugging, and code optimization.
- Guided learners through core algorithmic techniques including divide-and-conquer, dynamic programming, greedy algorithms, and graph traversal.
- Reviewed student code submissions to improve efficiency, correctness, and adherence to asymptotic complexity targets.

Student Technician, IT Department, Caldwell University **July 2024 – Aug 2025**

- Supported campus-wide IT systems, ensuring reliable day-to-day technology operations for students, faculty, and staff.
- Assisted users with troubleshooting university portal access, account issues, and hardware/software problems.
- Rebuilt and configured computers, optimizing system performance and reliability across user workstations.
- Collaborated with IT staff to maintain hardware, software, and network integrity across the campus environment.

Media Assistant, Caldwell University **Jan 2024 – Aug 2025**

- Managed audio/visual setup, operation, and teardown for campus events, lectures, and livestreams, ensuring seamless production from start to finish.
- Operate and maintain media equipment — cameras, microphones, mixing boards, and video-conferencing systems — and troubleshoot issues in real time during live events.
- Produce and edit multimedia content in support of university communications, coordinating closely with faculty, staff, and event organizers.

Peer Tutor, Academic Success Center, Caldwell University **Sept 2024 – Present**

- Tutored 25+ students in Python and algorithms, strengthening their computational problem-solving skills.
- Guiding learners through topics in web development, discrete math, and algorithmic logic.
- Helping students apply programming concepts to data analysis tasks and efficient code design.

Campus Store Team Member — Barnes & Noble, Caldwell University **July 2024 – May 2026**

- Analyzed textbook inventory, enrollment data, and sales trends to forecast demand and maintain accurate stock levels.
 - Used POS and inventory systems to ensure accurate data entry and support data-driven operational decisions.
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SKILLS

- **Languages:** Python, R, JavaScript, C, SQL (MySQL), HTML/CSS
 - **Libraries & Frameworks:** PyTorch, TensorFlow, Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, Flask, NLTK, shiny
 - **Tools:** Git/GitHub, Jupyter Notebook, VS Code, Excel, Word
 - **Domains:** Machine Learning, Deep Learning, NLP, Data Analysis, Data Visualization, Research
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PROJECTS

Next Word Predictor | *PyTorch, LSTM, NLTK* | [GitHub](#)

- Designed and implemented an LSTM-based next-word prediction model using PyTorch with an embedding → LSTM → linear architecture.
- Built a complete NLP pipeline including NLTK tokenization, vocabulary construction (249 tokens), sequence generation (631 sequences), and padded batching (max length: 127).
- Trained the model for 50 epochs (batch size: 32) using Adam optimizer and cross-entropy loss, achieving **99.52% next-word prediction accuracy** with final loss of 1.35.
- Implemented both single-step and iterative text generation for autocomplete-style sequence completion.

Video Content Retrieval-Augmented Generation (RAG) System | [GitHub](#)

- Engineered end-to-end machine learning pipelines using Python, pandas, and scikit-learn for automated data preprocessing, feature extraction, and model training.
- Implemented NLP workflows integrating OpenAI Whisper and BGE-M3 embeddings for semantic text analysis and context understanding.
- Developed retrieval-augmented generation (RAG) models leveraging vector embeddings and cosine similarity for context-aware information retrieval.
- Automated data ingestion, transformation, and parallel processing for JSON and multimedia datasets using pandas, NumPy, and Joblib to optimize pipeline efficiency.